

**SRI CHANDRASEKHARENDRA SARASWATHI
VISWA MAHAVIDYALAYA**

(UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956)

ENATHUR, KANCHIPURAM – 631 561

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



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Course Code : 2CSEC03

Course Name : Human and Computer Interaction

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Bonafide Certificate

This is to certify that this is the Bonafide record of work done by **Mr./Ms. Akella Lakshmi Thanmai**, with Reg No. **112547001** of I- M.E(CSE) during the academic year 2025-2026.

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Experiment - I

Title: Study of Human I/O Channels and Memory

Objectives: To understand human perceptual channels (visual, auditory, haptic) and memory types through task analysis.

Task: Analyse 3 real-world interfaces (ATM, elevator panel, mobile lock screen). Identify input/output channels used. Map short-term and long-term memory usage.

Theory:

Human-Computer Interaction (HCI) is the study of how humans interact with computer systems and how systems can be designed to support effective, efficient, and satisfying interaction. Understanding human perceptual capabilities and memory limitations is essential for designing user-friendly interfaces.

Human beings interact with computers through different sensory channels, mainly visual, auditory, and haptic channels. These channels help users receive information from the system and provide input to perform tasks. The effectiveness of an interface depends on how well it utilizes these human input and output channels.

Human Input and Output Channels

Visual Channel

The visual channel is the most dominant sensory channel in human interaction. It involves the use of eyes to perceive text, images, icons, colors, and layout on screens. Most computer interfaces rely heavily on visual output to communicate information. Good visual design improves readability, recognition, and understanding while reducing cognitive load.

Auditory Channel

The auditory channel involves hearing sounds such as alerts, notifications, voice prompts, and feedback tones. It is useful when visual attention is limited or when immediate feedback is required. Auditory signals help users detect errors and system status changes quickly.

Haptic Channel

The haptic channel involves touch and physical interaction with devices such as keyboards, mouse, touchscreens, and vibration feedback. It helps users confirm actions and improves interaction accuracy through tactile feedback.

Human Memory

Memory plays a crucial role in how users interact with computer systems. Designers must understand memory limitations to create usable interfaces.

Short-Term Memory (STM)

Short-term memory temporarily stores information for a short duration, typically a few seconds. It has limited capacity, usually about 7 ± 2 items. Interfaces that require users to

remember many steps or large amounts of information increase cognitive load and may lead to errors.

Examples include remembering an OTP, temporary instructions, or menu options.

Long-Term Memory (LTM)

Long-term memory stores information for extended periods, from hours to a lifetime. It includes learned skills, knowledge, and experiences. Interfaces that rely on familiar concepts and consistent design help users rely on long-term memory, making interaction easier.

Examples include remembering passwords, PINs, or frequently used procedures.

Importance of Studying Human I/O Channels and Memory

Understanding human perceptual channels and memory limitations helps designers create systems that:

- Reduce user errors
- Improve efficiency
- Minimize mental workload
- Enhance usability and user satisfaction
- Provide effective feedback

By designing interfaces that align with human capabilities, systems become easier to learn and use.

Application in Interface Design

Real-world interfaces such as ATM machines, elevator panels, and mobile lock screens use visual displays, auditory feedback, and touch input while minimizing memory load through clear instructions and simple interaction steps.

Studying these interfaces helps in understanding how human cognitive limits influence interface design and usability.

Conclusion

Thus, the experiment helped us understand the role of human input and output channels such as visual, auditory, and haptic in interaction with computer systems. It also highlighted the importance and limitations of short-term and long-term memory in interface usage. Understanding these factors helps in designing more efficient, user-friendly, and cognitively supportive systems.

Observation:

1. The user depended mainly on visual cues such as menus and labels.
2. Audio alerts helped the user notice important system events.
3. Touch interaction was used to navigate through options.
4. The user needed to remember instructions briefly during the task.
5. The user recalled previous experience while interacting with the system.
6. The user recognized icons instead of remembering commands.
7. The user relied on screen prompts to complete actions.
8. Visual design helped reduce confusion during interaction.
9. The user performed tasks smoothly using available input channels.
10. Memory usage was minimal due to clear interface guidance.

Experiment - II

Title: Ergonomic Evaluation of Computer Workstation

Objectives: To analyze ergonomics and physical interaction factors..

Task: Measure posture, keyboard, mouse, and screen placement.

- Identify ergonomic violations.
- Propose design improvements.

Theory

Ergonomics is the study of designing workplaces, systems, and products to fit the physical and psychological capabilities of users. In Human–Computer Interaction (HCI), ergonomics focuses on creating computer workstations that ensure comfort, safety, efficiency, and reduced physical strain during interaction.

A computer workstation typically consists of a chair, desk, monitor, keyboard, mouse, and other input devices. Improper arrangement of these components may lead to physical discomfort, fatigue, musculoskeletal disorders, eye strain, and long-term health issues. Therefore, ergonomic evaluation is essential to improve user well-being and productivity.

Posture and Body Alignment

Proper posture is a fundamental ergonomic principle. The user should sit upright with:

- Back supported by the chair
- Feet flat on the floor
- Knees at approximately 90 degrees
- Shoulders relaxed
- Wrists straight while typing

Incorrect posture such as slouching, bending forward, or twisting can cause back pain, neck strain, and repetitive stress injuries.

Monitor Placement

The monitor should be placed:

- At eye level or slightly below
- Approximately 50–70 cm away from the user
- Directly in front of the user to avoid neck rotation

Improper monitor placement may lead to eye strain, headaches, and neck pain.

Keyboard and Mouse Placement

The keyboard and mouse should be positioned:

- At elbow height
- Close to the user
- With wrists in a neutral (straight) position

If the keyboard is too high or too far, it may cause wrist strain and increase the risk of repetitive strain injuries such as carpal tunnel syndrome.

Lighting and Environment

Adequate lighting is important to reduce glare and eye fatigue. The workstation should avoid direct light reflections on the screen. Proper ventilation and noise control also contribute to a comfortable working environment.

Importance of Ergonomic Evaluation

Evaluating ergonomics helps to:

- Prevent physical injuries
- Reduce fatigue and discomfort
- Improve work efficiency
- Increase user satisfaction
- Promote long-term health

In HCI, ergonomic design ensures that technology adapts to human capabilities rather than forcing users to adjust to poorly designed systems.

Conclusion:

Thus, the ergonomic evaluation of the computer workstation helped us understand the importance of proper posture and correct placement of the monitor, keyboard, and mouse. Identifying ergonomic violations allows us to suggest improvements that reduce physical strain, prevent injuries, and enhance user comfort and productivity.

Observation:

1. The user leaned slightly forward while working.
2. The monitor was placed slightly below eye level.
3. The keyboard position required the user to bend the wrists slightly.
4. The mouse usage caused mild wrist strain.
5. The user adjusted posture frequently during the task.
6. The screen brightness affected visual comfort slightly.
7. The user experienced slight neck discomfort after some time.
8. The chair lacked proper back support.
9. The user felt some fatigue after continuous work.
10. Improved workstation arrangement could increase comfort.

Experiment - III

Title: Interaction Paradigms Analysis

Objectives: To compare different interaction paradigms.

Task: Study CLI, GUI, Touch-based, Voice-based interfaces.

- Compare learnability, efficiency, and error rate.

Theory

Interaction paradigms refer to the different ways in which users communicate and interact with computer systems. In Human–Computer Interaction (HCI), understanding various interaction paradigms helps designers create systems that are efficient, user-friendly, and suitable for different types of users.

An interaction paradigm defines the style of communication between the user and the system. Different paradigms are developed based on technological advancements and user needs. The most common paradigms include Command Line Interface (CLI), Graphical User Interface (GUI), Touch-based Interface, and Voice-based Interface.

Command Line Interface (CLI)

A Command Line Interface allows users to interact with the system by typing textual commands using a keyboard. The system processes the command and provides text-based output.

CLI requires users to remember commands and syntax, which increases cognitive load. However, it is highly efficient for experienced users and provides greater control over the system.

Graphical User Interface (GUI)

A Graphical User Interface enables users to interact with the system through visual elements such as icons, menus, buttons, and windows. Users typically use a mouse or touchpad along with a keyboard.

GUI is easy to learn because it supports recognition rather than recall. It reduces user errors and improves usability, making it suitable for general users.

Touch-Based Interface

Touch-based interfaces allow users to interact directly with the system using gestures such as tapping, swiping, and pinching on a touchscreen device. This paradigm is widely used in smartphones and tablets.

Touch interfaces are intuitive and easy to learn, but small screen sizes may sometimes increase input errors.

Voice-Based Interface

Voice-based interfaces enable users to interact with systems using spoken commands. The system processes speech input and responds through voice or visual output.

This paradigm provides hands-free interaction and accessibility support. However, it may be affected by background noise, pronunciation differences, and speech recognition errors.

Comparison of Interaction Paradigms

Interaction paradigms can be compared based on:

- **Learnability:** How easily new users can learn the system
- **Efficiency:** Speed and performance in completing tasks
- **Error Rate:** Likelihood of user mistakes

CLI is efficient for experts but difficult for beginners. GUI and touch-based interfaces are easier to learn and have lower error rates. Voice-based interfaces provide accessibility but may face recognition challenges.

Importance of Studying Interaction Paradigms

Studying different interaction paradigms helps designers:

- Choose the appropriate interaction style
- Improve usability and accessibility
- Reduce cognitive load
- Enhance user satisfaction

Understanding interaction models ensures that systems are designed according to user needs and context of use.

Conclusion

Thus, the study of different interaction paradigms such as CLI, GUI, touch-based, and voice-based interfaces helped us understand their advantages and limitations. By comparing them based on learnability, efficiency, and error rate, we learned that different paradigms are suitable for different users and contexts. Understanding these models supports better interface design and improved user experience.

Observation:

1. The user found CLI interfaces difficult to learn.
2. Commands had to be remembered before use.
3. Typing mistakes caused errors in command execution.
4. The user took more time to perform tasks in CLI.
5. Voice interaction was convenient for simple commands.
6. Background noise sometimes affected voice recognition.
7. The user preferred GUI for most activities.
8. CLI was useful for advanced users with technical knowledge.
9. Different paradigms offered different advantages.
10. User performance depended on familiarity with the interface.

Experiment - IV

Title: Scenario-Based Interaction Design

Objectives: To design user scenarios and personas.

Task: Create personas for a library management system or food delivery app.

- Develop usage scenarios and task flows.

Theory

Scenario-Based Interaction Design is a user-centred design approach in Human–Computer Interaction (HCI) that focuses on understanding users, their goals, and how they interact with a system in real-life situations. Instead of starting with technical features, this method begins by analyzing users and their needs.

The core idea of scenario-based design is to create realistic descriptions of how users will use a system. These descriptions help designers visualize interactions and identify usability issues before actual system development.

Personas

A persona is a fictional representation of a real user group. It includes details such as:

- Name and age
- Occupation
- Goals
- Needs
- Problems
- Technical skills

Personas help designers understand different types of users and ensure the system is designed according to their characteristics and expectations.

Usage Scenarios

A usage scenario is a short story describing how a persona interacts with the system to achieve a specific goal. It explains:

- What the user wants to do
- How they perform the task
- What challenges they face

Scenarios help designers focus on real-world situations and improve usability.

Task Flow

Task flow represents the step-by-step sequence of actions a user performs to complete a task. It shows how users move from one screen or action to another.

Designing proper task flows ensures:

- Logical navigation
- Reduced complexity
- Fewer user errors
- Better efficiency

Importance of Scenario-Based Design

Scenario-based design helps in:

- Understanding user needs clearly
- Designing user-friendly systems
- Reducing usability problems
- Supporting user-centred design thinking
- Improving overall user experience

It ensures that systems are built around users rather than forcing users to adapt to the system.

Conclusion

Thus, the Scenario-Based Interaction Design experiment helped us understand the importance of creating personas, usage scenarios, and task flows in designing user-centred systems. This approach ensures that systems are developed according to real user needs, improving usability and overall user experience.

Observation:

1. The user read the scenario to understand the task.
2. The task flow helped guide the user through the process.
3. Some steps required extra explanation.
4. The user interacted with the system according to the scenario.
5. The design supported the user's objectives.
6. The user completed the activity step by step.
7. The scenario provided a clear understanding of system usage.
8. The interaction process was logical.
9. Minor improvements could simplify some steps.
10. Overall, the system supported user tasks effectively.

Experiment - V

Title: Low-Fidelity Prototyping

Objectives: To create paper prototypes.

Task: Design a paper prototype for a mobile or web application. Perform usability walkthrough with peers.

Theory:

Low-Fidelity Prototyping is a design technique in Human–Computer Interaction (HCI) used to create simple and rough representations of a system before actual development. These prototypes are usually made using paper, sketches, or basic digital drawings and focus on layout, structure, and interaction flow rather than detailed visuals or functionality.

A low-fidelity prototype represents the basic structure of an application, including screens, buttons, menus, and navigation flow. It helps designers visualize how users will interact with the system and identify usability problems at an early stage.

Unlike high-fidelity prototypes, low-fidelity prototypes do not include detailed graphics, colors, animations, or real data. The main purpose is to test ideas quickly and economically.

Paper Prototyping

Paper prototyping is a common low-fidelity method where designers draw interface screens on paper. Each screen is represented separately, and interactions are simulated manually. When a user selects an option, the corresponding paper screen is shown.

This method allows:

- Quick modification of design
- Easy feedback collection
- Cost-effective testing
- Early detection of usability issues

Usability Walkthrough

A usability walkthrough involves asking users or peers to perform tasks using the prototype. The observer notes:

- Confusion or hesitation
- Errors made
- Navigation difficulties
- User feedback

This process helps in improving the design before actual implementation.

Importance of Low-Fidelity Prototyping

Low-fidelity prototyping helps in:

- Early validation of design ideas
- Reducing development cost
- Improving user-centred design
- Encouraging creative thinking
- Identifying navigation and layout issues

It supports iterative design, where improvements are made continuously based on user feedback.

Conclusion

Thus, the Low-Fidelity Prototyping experiment helped us understand the importance of creating simple paper prototypes to test design ideas at an early stage. It enables quick feedback, identifies usability issues, and supports user-centred design before system development.

Observation:

1. The user examined the paper prototype carefully.
2. The interface layout was understandable.
3. Some buttons required clearer placement.
4. The prototype helped visualize navigation steps.
5. The user suggested improving screen organization.
6. The user identified areas needing design changes.
7. The prototype supported early design testing.
8. Interaction flow became easier to understand.
9. The user found the design simple.
10. Feedback helped improve the design concept.

Experiment – VI

Title: Navigation and Screen Design.

Objective: To design effective navigation and screen layouts.

Task: Design wireframes using tools like Figma/Balsamiq.

- Apply design principles and guidelines.

Theory:

Navigation and screen design are important aspects of Human–Computer Interaction (HCI) that focus on how users move through a system and how information is presented on the screen. Effective navigation and well-organized screen layouts help users perform tasks easily, efficiently, and without confusion.

Navigation refers to the structure that allows users to move between different pages or screens of an application. It includes elements such as menus, buttons, links, tabs, and icons. A good navigation system helps users understand where they are in the system and how to move to other sections.

Screen design focuses on the arrangement of interface elements such as text, images, buttons, input fields, and menus on the screen. Proper screen layout improves readability, reduces cognitive load, and enhances user experience.

Wireframes are commonly used to design navigation and screen layouts. A wireframe is a simple visual representation of an interface that shows the placement of elements and the structure of the screen without detailed graphics or colors. Tools like Figma and Balsamiq are widely used to create wireframes for web and mobile applications.

When designing screens and navigation, several HCI design principles should be followed, such as consistency, simplicity, visibility of important elements, clear feedback, and logical grouping of information. These principles ensure that the interface is easy to learn, easy to use, and reduces user errors.

Proper navigation and screen design play a significant role in improving usability, accessibility, and overall interaction between users and computer systems.

Conclusion:

Thus, the experiment helped us understand the importance of effective navigation and proper screen layout in interface design. By creating wireframes and applying HCI design principles, we can design user-friendly systems that improve usability and enhance the overall user experience.

Observation:

1. The user understood the navigation structure easily.
2. Menus helped the user move between screens.
3. Icons were clear and recognizable.
4. The user followed the navigation flow successfully.
5. Screen design supported task completion.
6. Some elements could be better grouped.
7. The user found the interface simple.
8. The layout helped reduce errors.
9. Navigation improved user control over the system.
10. The design supported effective interaction.

Experiment – VII

Title: Iterative Design & Prototyping

Objective: To understand iterative design.

Task: Improve an existing prototype based on usability feedback.

- Document design rationale.

Theory: Iterative Design is an important concept in Human–Computer Interaction (HCI) that focuses on continuously improving a system through repeated cycles of design, testing, evaluation, and refinement. Instead of designing a system once and implementing it directly, iterative design allows designers to gradually improve the interface based on user feedback and usability testing.

In this approach, a prototype of the system is first created to represent the initial design idea. A prototype may be a simple paper design, wireframe, or digital interface model that demonstrates how the system will function. The prototype is then evaluated by users to identify usability issues, errors, or difficulties in interaction.

Based on the feedback collected during evaluation, designers make necessary improvements to the prototype. This process of designing, testing, and modifying the system is repeated several times until the interface becomes more efficient, user-friendly, and effective.

Iterative design helps designers understand user needs better and reduces the risk of developing systems that are difficult to use. It supports user-centred design by involving users throughout the design process.

Design Rationale

Design rationale refers to documenting the reasons behind design decisions. It explains why certain design choices were made, what alternatives were considered, and how the final design improves usability.

Documenting design rationale helps designers:

- Understand the decision-making process
- Maintain consistency in design
- Improve future design iterations
- Communicate design ideas clearly within the team

Importance of Iterative Design

Iterative design is important because it:

- Improves usability and user experience

- Allows early detection of design problems
- Encourages continuous improvement
- Reduces development risks and errors
- Supports user-centred system development

Through repeated testing and refinement, systems become more effective and easier to use.

Conclusion:

Thus, the Iterative Design and Prototyping experiment helped us understand the importance of improving system designs through repeated testing and user feedback. Iterative design ensures better usability, while documenting design rationale helps explain design decisions and supports effective system development.

Observation:

1. The updated design solved previous usability issues.
2. The user could complete tasks faster.
3. The improved layout reduced confusion.
4. Navigation became clearer after modifications.
5. The user understood the interface easily.
6. Feedback was successfully applied in the design.
7. The user felt more comfortable using the system.
8. Errors decreased compared to earlier design.
9. The interaction process became smoother.
10. Iterative design improved the final prototype.

Experiment – VIII

Title: Cognitive Model Analysis (GOMS / KLM)

Objective: To analyse user task performance.

Task: Apply GOMS or the Keystroke Level Model to compare two interfaces.

- Calculate task completion time.

Theory: Cognitive models in Human–Computer Interaction (HCI) are used to analyze how users interact with computer systems and how efficiently they perform tasks. These models help designers understand the mental processes involved when users complete actions in an interface. Cognitive model analysis allows designers to predict user performance and identify ways to improve interface efficiency.

Two widely used cognitive models are GOMS and the Keystroke Level Model (KLM). These models help evaluate the time required to complete tasks and compare different interface designs.

GOMS Model

GOMS stands for:

- **G – Goals:** What the user wants to achieve
- **O – Operators:** Basic actions required to complete the task (click, type, move mouse)
- **M – Methods:** Different ways to perform the task
- **S – Selection Rules:** Rules that determine which method the user chooses

The GOMS model breaks down a task into smaller steps and analyzes how users perform those steps. It helps in understanding how users accomplish tasks and how interface design affects task efficiency.

Keystroke Level Model (KLM)

The Keystroke Level Model is a simplified version of GOMS used to estimate the time taken by an expert user to complete a task without errors. It measures the time required for basic actions such as:

- **K (Keystroke):** Pressing a key or button
- **P (Pointing):** Moving the mouse to a target
- **H (Homing):** Moving hands between keyboard and mouse
- **M (Mental Preparation):** Time needed to think before an action
- **R (Response Time):** System response delay

By adding the time required for these actions, designers can estimate the total time needed to complete a task.

Importance of Cognitive Model Analysis

Cognitive models are useful because they:

- Predict user performance
- Compare different interface designs
- Identify inefficiencies in interaction
- Improve usability and efficiency
- Support data-driven interface design decisions

Using GOMS and KLM, designers can evaluate how quickly and efficiently users can complete tasks and improve the interaction process.

Conclusion:

Thus, the cognitive model analysis experiment helped us understand how user task performance can be evaluated using models like GOMS and KLM. These models allow designers to estimate task completion time and compare different interfaces, helping to improve interaction efficiency and usability.

Observation:

1. The user followed a sequence of operations to complete the task.
2. Some actions required additional time.
3. Mental preparation increased task completion time.
4. The interface required multiple interaction steps.
5. The user needed to move between keyboard and mouse frequently.
6. The user took slightly longer to complete the task.
7. The interface complexity affected performance.
8. The user experienced small delays in task execution.
9. Fewer steps could improve efficiency.
10. The task completion time was higher compared to the other interface.

Experiment – IX

Title: Socio-Organisational Stakeholder Analysis

Objective: To identify stakeholder requirements.

Task: Identify stakeholders in a healthcare or education system. Map functional and non-functional requirements.

Theory: Socio-Organisational Stakeholder Analysis is used in Human-Computer Interaction (HCI) to identify all people who are directly or indirectly affected by a system. These people are called stakeholders. Stakeholders may include users, administrators, managers, technicians, and support staff.

In any system such as healthcare or education, different stakeholders have different needs and expectations. For example, in a healthcare system, doctors need quick access to patient data, nurses need easy data entry, and administrators require reports. Similarly, in an education system, students, teachers, and administrators interact with the system differently.

Stakeholder analysis helps designers understand functional and non-functional requirements. Functional requirements describe what the system should do, such as storing records, generating reports, and managing users. Non-functional requirements describe how the system should perform, such as usability, reliability, security, and performance.

By identifying stakeholder needs, designers can develop systems that support organizational workflow, improve communication, and enhance efficiency. This ensures that the system is useful for all users and supports the overall organizational goals.

Conclusion:

The Socio-Organisational Stakeholder Analysis helped identify different stakeholders and their requirements. Understanding both functional and non-functional needs improves system usability and supports effective organizational design.

Observation:

1. The user identified different stakeholders in the system.
2. The user recognized primary users such as students or doctors.
3. Secondary users such as administrators were also considered.
4. The user identified different requirements for each stakeholder.
5. Functional needs like data entry and record viewing were noted.
6. Non-functional needs like security and usability were observed.
7. The user understood the interaction between stakeholders.
8. The system supported multiple user roles.
9. The user observed the importance of workflow coordination.
10. Stakeholder analysis helped understand system design better.

Experiment – X

Title: Hypertext and Multimedia Interface Design

Objective: To design a hypermedia-based interface.

Task: Design a website structure using hypertext principles. Apply multimedia elements effectively.

Theory: Hypertext and multimedia interface design is an important concept in Human Computer Interaction (HCI). It focuses on designing interactive systems where users can navigate through linked information and multimedia content such as text, images, audio, and video.

Hypertext refers to text that contains links to other related information. These links allow users to move from one page to another in a non-linear way. Unlike traditional documents, hypertext allows flexible navigation based on user interest.

Multimedia refers to the use of multiple media elements such as:

- Text
- Images
- Audio
- Video
- Animation
- Graphics

Combining hypertext and multimedia creates a hypermedia interface. This type of interface is commonly used in websites, e-learning systems, and mobile applications.

In hypertext-based design, information is organized into pages connected by links. Users can navigate using menus, buttons, and hyperlinks. This improves usability and allows users to access information quickly.

Multimedia elements improve user understanding and make the interface more interactive. Images help visualization, audio provides instructions, and video demonstrates processes. However, multimedia should be used properly to avoid clutter and confusion.

Designers must follow navigation principles such as:

- Clear menu structure
- Consistent navigation
- Logical page hierarchy
- Proper labeling of links

- Easy back and home options

A well-designed hypermedia interface improves user experience and information accessibility.

Example

Website Structure (College Website)

Home

→About

→Departments

→Courses

→Faculty

→ Contact

Multimedia Used:

- Images for departments
- Video for campus tour
- Audio for announcements
- Icons for navigation

Outcome

- Understanding hypertext navigation
- Learning multimedia usage
- Designing website structure
- Improving information architecture
- Better user interaction design

Conclusion:

The hypertext and multimedia interface design experiment helped in understanding how linked information and multimedia elements improve navigation and user interaction. Proper use of hypermedia enhances usability and information accessibility.

Observation:

1. The user navigated through the interface using hyperlinks.
2. The menu structure helped the user move between pages easily.
3. Images improved understanding of the content.
4. Multimedia elements made the interface more interactive.
5. The user accessed different sections using navigation links.
6. The layout of the website was clear and organized.
7. The user followed the hierarchical structure of pages.
8. Icons and buttons helped in quick navigation.
9. The user found the interface easy to understand.
10. Hypertext links helped access information quickly.

Experiment – XI

Title: Mobile Application Interface Analysis.

Objective: To study mobile UI design.

Task: Analyze 3 mobile apps (game, utility, widget). Identify UI elements and interaction patterns.

Theory: Mobile Application Interface Analysis focuses on studying how mobile applications are designed for effective user interaction. In Human–Computer Interaction (HCI), mobile UI design is important because mobile devices have limited screen size, touch-based input, and performance constraints.

Mobile interfaces consist of various UI elements such as buttons, icons, menus, text fields, and navigation bars. These elements help users interact with the application easily. Since mobile devices rely mainly on touch input, interaction patterns such as tapping, swiping, scrolling, and pinching are commonly used.

Different types of mobile applications follow different design styles. Game applications focus on rich graphics and interactive controls, utility applications focus on simplicity and efficiency, and widget applications provide quick access to information with minimal interaction.

Designing mobile interfaces requires considering constraints such as small screen size, limited input methods, and the need for fast and responsive performance. A well-designed mobile interface should be simple, clear, and easy to use, ensuring a smooth user experience.

Analyzing mobile applications helps understand how UI elements and interaction patterns affect usability and user satisfaction.

Conclusion:

Thus, the experiment helped in understanding mobile UI design by analyzing different types of applications. It highlighted the importance of simple layouts, touch-based interaction, and efficient design to improve usability in mobile systems.

Observation:

1. The user interacted with the mobile apps using touch gestures.
2. The user used tapping and swiping for navigation.
3. The game app provided interactive and engaging controls.
4. The utility app was simple and easy to use.
5. The widget app provided quick access to information.
6. Icons and buttons were clearly visible on the screen.
7. The user found the interface responsive and fast.
8. The user completed tasks easily on the mobile apps.
9. The small screen required simple and clear design.
10. The overall interaction experience was smooth.

Experiment – XII

Title: Mobile Information Architecture Design

Objective: To design mobile navigation.

Task: Create sitemap and flow diagrams for a mobile app. Apply Mobile 2.0 principles.

Theory: Mobile Information Architecture Design is a key concept in Human–Computer Interaction (HCI) that focuses on organizing content and navigation in mobile applications. It ensures that users can easily access information and complete tasks efficiently.

A sitemap is a hierarchical representation of all screens or pages in the application. It helps designers understand how content is structured. A flow diagram represents the sequence of actions and navigation paths followed by the user while interacting with the app.

Mobile-first design is an approach where applications are designed primarily for mobile devices before adapting to larger screens. It emphasizes simplicity, clarity, and efficiency.

Mobile 2.0 principles focus on:

- Simplicity in design
- Minimal user effort
- Fast navigation
- Touch-friendly interfaces
- Clear content hierarchy

A well-designed mobile information architecture improves usability, reduces confusion, and enhances user experience.

Example

Sitemap (Food Delivery App)

Home

Search

Categories

Restaurant List

Food Details

Cart

Payment

Order Tracking

Flow Diagram

Open App → Search Food → Select Item → Add to Cart → Payment → Order Confirmation

Conclusion:

Thus, the experiment helped in understanding how to design mobile information architecture using sitemaps and flow diagrams. Applying mobile-first principles ensures simple navigation, better usability, and improved user experience.

Observation:

1. The user followed the sitemap structure to understand the app layout.
2. The navigation between screens was simple and clear.
3. The user easily moved from one screen to another.
4. The flow diagram helped in understanding task steps.
5. The user found the app structure well organized.
6. The user completed tasks with minimal effort.
7. The user understood the sequence of actions clearly.
8. The navigation reduced confusion during interaction.
9. The design supported easy access to information.
10. The user experienced smooth navigation.

Experiment – XIII

Title: Mobile UI Prototype

Objective: To build a mobile app prototype.

Task: Design interactive prototype using Figma / Adobe XD. Include gestures, transitions, feedback.

Theory: Mobile UI Prototyping is an important concept in Human–Computer Interaction (HCI) that focuses on designing and testing mobile interfaces before implementation. It allows designers to simulate user interaction and evaluate usability.

An interactive prototype includes:

1. Gestures

- Tap
- Swipe
- Scroll
- Long press

These are the main ways users interact with mobile apps.

2. Transitions

Transitions define how one screen moves to another, such as:

- Slide
- Fade
- Pop-up

They improve the visual flow and user experience.

3. Feedback

Feedback informs the user about system response:

- Button highlight
- Loading indicator
- Success message

Good feedback ensures users know their actions are recognized.

Importance of Mobile UI Prototyping

- Helps visualize app design
- Detects usability issues early
- Saves development time and cost
- Improves user experience
- Supports iterative design

Prototyping allows designers to test ideas and refine them before final development.

Example

Prototype Screens (Food Delivery App)

- Login Screen
- Home Screen
- Search Screen
- Food Details
- Cart
- Payment

Interaction Flow

Login → Home → Search → Select Item → Add to Cart → Payment

Features

- Tap to select items
- Swipe to scroll
- Transition between screens
- Feedback on button click

Conclusion:

Thus, the experiment helped in understanding how to build interactive mobile UI prototypes using design tools. It highlights the importance of gestures, transitions, and feedback in creating user-friendly mobile applications.

Observation:

1. The user interacted with the prototype using touch gestures.
2. The user used tapping to navigate between screens.
3. Transitions made navigation smooth and clear.
4. Feedback confirmed user actions effectively.
5. The user easily understood the app flow.
6. The prototype simulated real app experience.
7. The user completed tasks without confusion.
8. The interface was simple and user-friendly.
9. The user found navigation responsive.
10. The overall interaction was smooth.

Experiment – XIV

Title: Designing Web Interfaces with Interaction Techniques

Objective: To implement modern web interactions.

Task: Design a web interface demonstrating: Drag & Drop

Direct Selection

Contextual tools

Overlays and Inlays

Theory: Designing web interfaces with interaction techniques is an important concept in Human–Computer Interaction (HCI). It focuses on improving the way users interact with web systems by using advanced interaction methods.

Modern web design emphasizes:

- Direct manipulation of objects
- Reduced user effort
- Faster task completion
- Better visual feedback

Interaction techniques like drag & drop and direct selection allow users to interact naturally with the interface. Contextual tools reduce screen clutter by showing options only when needed. Overlays and inlays help present additional information without navigating away from the current page.

These techniques follow important design principles such as:

- Visibility
- Feedback
- Consistency
- Efficiency

Proper implementation of these techniques enhances usability and improves overall user experience.

Example

Web Interface (File Manager)

- Drag files into folders
- Select items directly

- Right-click to open contextual menu
- Use pop-up window for file details

Outcome

- Understanding modern interaction techniques
- Designing interactive web interfaces
- Improving usability and efficiency
- Enhancing user experience

Conclusion

Thus, the experiment helped in understanding modern web interaction techniques such as drag & drop, direct selection, contextual tools, and overlays. These techniques improve usability, reduce user effort, and enhance the overall user experience.

Observation:

1. The user used drag and drop to move objects easily.
2. Direct selection helped the user interact quickly with elements.
3. Contextual tools appeared when needed.
4. Overlays displayed additional information clearly.
5. Inlays expanded content within the page.
6. The user found interaction smooth and intuitive.
7. Tasks were completed faster using these techniques.
8. The interface reduced the need for complex steps.
9. The user received proper feedback during actions.
10. Overall interaction was efficient and user-friendly.

Experiment – XV

Title: Process Flow Design

Objective: To design task-based workflows.

Task: Create flow diagrams for e-commerce checkout or registration process.

Identify usability bottlenecks.

Theory: Process Flow Design is a fundamental concept in Human–Computer Interaction (HCI) that focuses on structuring user tasks into a clear and logical sequence. It represents how users interact with a system to achieve a specific goal, such as purchasing a product, registering for an account, or booking a service.

A process flow is typically represented using a flow diagram, which visually shows the sequence of steps, decisions, and actions involved in completing a task. These diagrams include elements such as start points, process steps, decision points, and end points. Flow diagrams help designers understand how users move through a system and identify areas that may cause confusion or inefficiency.

In HCI, designing effective process flows is essential to ensure usability. A well-designed flow should be simple, intuitive, and efficient. Users should be able to complete tasks with minimal effort and without unnecessary steps. The flow should guide users clearly from one step to the next, ensuring that they always know what to do.

One of the key aspects of process flow design is identifying usability bottlenecks. Bottlenecks are points in the workflow where users face difficulty, delay, or confusion. These may occur due to poor design decisions such as complex navigation, unclear instructions, too many input fields, or lack of feedback. Identifying and removing these bottlenecks improves user experience and increases task completion rates.

Process flow design also follows important design principles such as:

- Simplicity – Keeping the process easy to understand
- Consistency – Maintaining uniform design across steps
- Visibility – Showing important information clearly
- Feedback – Informing users about their actions
- Efficiency – Reducing unnecessary steps

In modern systems, process flow design plays a critical role in improving user satisfaction and system performance. A good workflow ensures smooth interaction, reduces errors, and enhances overall usability.

Conclusion:

Thus, the process flow design experiment helped in understanding how to organize user tasks into a clear and logical sequence. It highlighted the importance of designing workflows that are simple, efficient, and easy to follow. Identifying usability bottlenecks helped in improving the overall performance of the system. The experiment also showed how proper design reduces user errors and confusion. Overall, it enhanced understanding of process-oriented design and its role in improving user experience.

Observation:

1. The user followed the workflow step by step.
2. The flow diagram made the process easy to understand.
3. The user completed the task without confusion.
4. Each step was clearly defined.
5. Navigation between steps was smooth.
6. The user experienced minimal errors.
7. The process was logical and well-structured.
8. The user found the system easy to use.
9. Feedback was provided at each step.
10. The overall experience was efficient.

Experiment – XVI

Title: Case Study Analysis.

Objective: To evaluate real-world HCI systems.

Task: Select a popular system (Google Maps, Amazon, Uber).

Analyze usability, interaction styles, and design principles.

Theory: Case Study Analysis in Human–Computer Interaction (HCI) is the process of examining real-world systems to understand how effectively they are designed and how users interact with them. It helps in evaluating usability, interaction techniques, and adherence to design principles.

In HCI, usability is a key factor that determines how easily users can perform tasks using a system. A system with good usability ensures that users can complete tasks efficiently with minimal effort and errors. It includes factors such as learnability, efficiency, memorability, error rate, and user satisfaction.

Interaction styles play a significant role in user experience. Modern systems support multiple interaction styles such as touch, voice, and gesture-based interaction. These interaction methods allow users to communicate with the system in a natural and intuitive way.

Design principles guide the development of user interfaces. Principles such as consistency, visibility, feedback, and simplicity ensure that the interface is easy to understand and use. Consistency helps users predict system behavior, while feedback informs users about the results of their actions. Simplicity reduces cognitive load, and visibility ensures that important features are easily accessible.

Case study analysis also helps identify strengths and weaknesses in existing systems. By studying successful applications, designers can learn best practices and apply them to new systems. It also helps in understanding user needs and improving system performance.

Thus, case study analysis is an essential method in HCI for evaluating real-world systems and enhancing design quality.

Conclusion:

Thus, the case study analysis experiment helped in understanding how real-world systems are designed and used by people. It highlighted the importance of usability, interaction styles, and design principles in creating effective interfaces. The experiment also helped in identifying strengths and weaknesses of existing systems. By analyzing popular applications, we gained insights into good design practices. Overall, it improved our ability to evaluate and design user-friendly systems.

Observation:

1. The user found the system easy to navigate.
2. The interface was clear and well-organized.
3. The user completed tasks quickly.
4. Feedback was provided after each action.
5. The design was consistent throughout.
6. The user understood the interface easily.
7. The system responded quickly to inputs.
8. The user found features useful.
9. Interaction was smooth and efficient.
10. Overall user experience was good.

Experiment – XVII

Title: Speech-Based Interface Design.

Objective: To understand voice interaction.

Task: Design dialog flow for a voice assistant.

Identify challenges in speech recognition.

Theory: Speech-Based Interface Design is an advanced concept in Human–Computer Interaction (HCI) that enables users to interact with systems through spoken language. It replaces traditional input methods such as typing or clicking with natural voice communication, making interaction more intuitive and accessible.

Speech-based systems use speech recognition technology to convert spoken words into text and then process them to perform specific actions. These systems also use natural language processing (NLP) to understand the meaning and intent behind user commands.

A key component of speech-based interaction is the dialog flow, which defines the structure of conversation between the user and the system. A well-designed dialog flow ensures smooth communication by guiding the user through a sequence of prompts, responses, and actions. It includes handling user input, providing system feedback, and managing errors effectively.

Speech-based interfaces offer several advantages. They enable hands-free interaction, making them useful in situations where manual input is not possible. They also improve accessibility for users with physical disabilities or limited mobility. Additionally, they provide a more natural way of interacting with technology.

However, there are several challenges in speech-based interaction. Speech recognition systems may struggle with variations in accents, pronunciation, and language. Background noise can also affect accuracy. Misinterpretation of commands can lead to incorrect system responses. Therefore, it is important to design systems that can handle errors gracefully and provide clear feedback to users.

Design principles such as simplicity, clarity, and feedback are essential in speech interface design. Commands should be easy to understand, and the system should respond clearly and accurately. Providing confirmation messages and error prompts helps users interact more effectively.

Thus, speech-based interface design plays a significant role in modern HCI by enabling natural and efficient communication between humans and machines.

Conclusion:

Thus, the speech-based interface design experiment helped in understanding how users interact with systems using voice commands. It highlighted the importance of designing clear and effective dialog flows for smooth communication. The experiment also helped in identifying challenges such as noise, accents, and misinterpretation in speech recognition. It improved

knowledge about accessibility and hands-free interaction. Overall, it provided a better understanding of conversational interface design in modern systems.

Observation:

1. The user interacted with the system using voice commands.
2. The system responded correctly to simple instructions.
3. The user found voice interaction easy and quick.
4. Some commands needed repetition.
5. The system provided audio feedback.
6. The user completed tasks hands-free.
7. Background noise slightly affected performance.
8. The user preferred simple commands.
9. The interaction was fast and convenient.
10. Overall experience was smooth.

Experiment – XVIII

Title: Multimodal Interaction Prototype.

Objective: To design a multimodal system.

Task: Design interface combining touch, voice, and visual feedback.

Document interaction flow.

Theory: Multimodal Interaction is an advanced concept in Human–Computer Interaction (HCI) that allows users to interact with systems using multiple modes of communication simultaneously. These modes include touch, voice, gestures, and visual feedback. Instead of relying on a single interaction method, multimodal systems combine different input and output techniques to improve usability and flexibility.

In traditional systems, users interact using only one method, such as typing or clicking. However, multimodal systems provide multiple options, allowing users to choose the most convenient way to interact. For example, a user can give a voice command while also using touch input to navigate the interface.

A key advantage of multimodal interaction is its ability to adapt to different user needs and environments. If one interaction method fails, another method can be used as an alternative. This improves reliability and ensures continuous interaction. For example, if voice input is not recognized due to noise, the user can switch to touch input.

Multimodal systems also improve accessibility. Users with disabilities can benefit from alternative input methods, making systems more inclusive. Additionally, combining multiple interaction modes reduces cognitive load and improves task efficiency.

Designing multimodal interfaces requires careful coordination between input methods and system responses. The system should provide clear and consistent feedback for all types of interaction. It is also important to ensure that different modes work together smoothly without causing confusion.

Common challenges in multimodal interaction include synchronization between modes, increased system complexity, and handling conflicting inputs. Designers must ensure that the system processes inputs accurately and responds appropriately.

Thus, multimodal interaction represents the future of interface design by providing more natural, flexible, and efficient ways for users to interact with technology.

Conclusion

Thus, the multimodal interaction prototype experiment helped in understanding how multiple interaction methods can be combined effectively. It highlighted the advantages of using touch, voice, and visual feedback together in a system. The experiment improved flexibility and efficiency in user interaction. It also emphasized the importance of accessibility and user

convenience. Overall, it provided knowledge about future-oriented interaction design in modern systems.

Observation:

1. The user interacted using both touch and voice inputs.
2. The system provided visual feedback for actions.
3. The user switched between input methods easily.
4. The interaction was flexible and smooth.
5. The user completed tasks efficiently.
6. The system responded correctly to inputs.
7. The user found the interface easy to use.
8. Feedback helped in understanding actions.
9. The interaction reduced user effort.
10. Overall experience was effective.